

Problem Set 75

Implicitly Complicated

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Problem 1

If V is the volume of a cube with edge length x and the cube expands as time passes, find $\frac{dV}{dt}$ in terms of $\frac{dx}{dt}$.

Given: x

Unknown: $\frac{dV}{dt}$

$$V = x^3$$

$$\frac{dV}{dt} = \boxed{3x^2 \frac{dx}{dt}}$$

Problem 2

Each side of a square is increasing at a rate of 6 cm/s. At what rate is the area of the square increasing when the area of the square is 16 cm²?

Given: $\frac{ds}{dt} = 6$ cm/s

Unknown: $\left. \frac{dA}{dt} \right|_{A=16}$

$$A = s^2$$

$$\frac{dA}{dt} = 2s \cdot \frac{ds}{dt} = 2 \cdot 4 \cdot 6 = \boxed{48 \text{ cm}^2/\text{s}}$$

The area of the square is increasing at a rate of 48 cm²/s.

Problem 4

Suppose $y = x^3 + 2x$, where x and y are functions of t . If $\frac{dx}{dt} = 5$, find $\frac{dy}{dt}$ when $x = 2$.

Given: $y = x^3 + 2x$, $\frac{dx}{dt} = 5$

Unknown: $\left. \frac{dy}{dt} \right|_{x=2}$

$$\frac{dy}{dt} = 3x^2 \cdot \frac{dx}{dt} + 2 \cdot \frac{dx}{dt} = 3 \cdot 4 \cdot 5 + 2 \cdot 5 = \boxed{70}$$

Problem 6

A plane flying horizontally at an altitude of 1 mi and a speed of 500 mi/h passes directly over a radar station. Find the rate at which the distance from the plane to the station is increasing when it is 2 mi away from the station.

Given: $h = 1$ mi, $\frac{dx}{dt} = 500$ mi/h

Unknown: $\frac{dy}{dt} \Big|_{x=2}$

$$\begin{aligned} y^2 &= h^2 + x^2 = 1 + x^2 \\ 2y \cdot \frac{dy}{dt} &= 2x \cdot \frac{dx}{dt} \\ \frac{dy}{dt} &= \frac{x}{\sqrt{1+x^2}} \cdot \frac{dx}{dt} = \frac{2}{\sqrt{5}} \cdot 500 = \boxed{200\sqrt{5} \text{ mi/h}} \end{aligned}$$

The distance from the plane to the station is increasing at a rate of $200\sqrt{5}$ mi/h.

Problem 8

Two cars start moving from the same point. One travels south at 60mi/h and the other travels west at 25mi/h. At what rate is the distance between the cars increasing two hours later?

Given: $\frac{dx}{dt} = 60$ mi/h, $\frac{dy}{dt} = 25$ mi/h

Unknown: $\frac{dz}{dt} \Big|_{t=2\text{h}}$

$$\begin{aligned} z^2 &= x^2 + y^2 \\ x &= \frac{dx}{dt} \cdot t = 120\text{mi} \\ y &= \frac{dy}{dt} \cdot t = 50\text{mi} \\ 2z \cdot \frac{dz}{dt} &= 2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} \\ \frac{dz}{dt} &= \frac{x \cdot \frac{dx}{dt} + y \cdot \frac{dy}{dt}}{\sqrt{x^2 + y^2}} = \frac{120 \cdot 60 + 25 \cdot 50}{130} = \boxed{65 \text{ mi/h}} \end{aligned}$$

The distance between the cars increases at a rate of 65 mi/h.

Problem 10

The altitude of a triangle is increasing at a rate of 1 cm/min while the area of the triangle is increasing at a rate of 2 cm²/min. At what rate is the base of the triangle changing when the altitude is 10 cm and the area is 100 cm²?

Given: $\frac{dh}{dt} = 1$ cm/min, $\frac{dA}{dt} = 2$ cm²/min.

Unknown: $\left. \frac{dB}{dt} \right|_{h=10, A=100}$

$$B = \frac{2A}{h}$$

$$\frac{dB}{dt} = \frac{2(h \cdot \frac{dA}{dt} - A \cdot \frac{dh}{dt})}{h^2} = \frac{2(10 \cdot 2 - 100 \cdot 1)}{100} = \boxed{-1.6 \text{ cm/min}}$$

The base of the triangle changes at a rate of -1.6 cm/min .

Problem 12

Water is leaking out of an inverted conical tank at a rate of $10,000 \text{ cm}^3/\text{min}$ at the same time that water is being pumped into the tank at a constant rate. The tank has height 6 m and the diameter at the top is 4 m . If the water level is rising at a rate of 20 cm/min when the height of the water is 2 m , find the rate at which water is being pumped into the tank.

Given: $h = 6 \text{ m}$, $\frac{dV_1}{dt} = -10000 \text{ cm}^3/\text{min}$, $\left. \frac{dh}{dt} \right|_{h=2} = 20$

Unknown: $\left. \frac{dV_2}{dt} \right|_{h=2}$

$$V = \frac{1}{3}\pi\left(\frac{h}{3}\right)^2 h = \frac{\pi h^3}{27}$$

$$\frac{dV_{net}}{dt} = \frac{\pi h^2}{9} \cdot \frac{dh}{dt} = \frac{\pi \cdot (200)^2 \cdot 20}{9} = \frac{800000\pi}{9} \approx 279252$$

$$\frac{dV_2}{dt} = \frac{dV_{net}}{dt} - \frac{dV_1}{dt} = 279252 - (-10000) = \boxed{289252 \text{ cm}^3/\text{min}}$$

Water is being pumped into the tank at a rate of $289252 \text{ cm}^3/\text{min}$.

Problem 14

Gravel is being dumped from a conveyor belt at a rate of $30 \text{ ft}^3/\text{min}$ and its coarseness is such that it forms a pile in the shape of a cone whose base diameter and height are always equal. How fast is the height of the pile increasing when the pile is 10 ft high?

Given: $\frac{dV}{dt} = 30 \text{ ft}^3/\text{min}$

Unknown: $\left. \frac{dx}{dt} \right|_{x=10}$

$$V = \frac{1}{3}\pi\left(\frac{x}{2}\right)^2 x = \frac{\pi x^3}{12}$$

$$\frac{dV}{dt} = \frac{\pi x^2}{4} \cdot \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{\frac{dV}{dt} \cdot 4}{\pi x^2} = \frac{30 \cdot 4}{\pi \cdot 10^2} = \boxed{\frac{5}{6\pi} \text{ ft/min}}$$

The height of the pile increases at a rate of $\frac{5}{6\pi} \text{ ft/min}$.

Problem 16

The Boyle's Law states that when a sample of gas is compressed at a constant temperature, the pressure P and the volume V satisfy the equation $PV = C$, where C is a constant. Suppose that at a certain instant the volume is 600 cm^3 , the pressure is 150 KPa , and the pressure is increasing at a rate of 20 KPa/min . At what rate is the volume decreasing at this instant?

Given: $\frac{dP}{dt} = 20 \text{ KPa/min}$

Unknown: $\frac{dV}{dt} \Big|_{V=600, P=150}$

$$V = \frac{C}{P}$$

$$\frac{dV}{dt} = C \cdot \left(-\frac{1}{P^2}\right) \cdot \frac{dP}{dt} = -\frac{V}{P} \cdot \frac{dP}{dt} = -\frac{600}{150} \cdot 20 = \boxed{-80 \text{ cm}^3/\text{min}}$$

The volume decreases at a rate of $-80 \text{ cm}^3/\text{min}$.

Problem 17

If two resistors with resistances R_1 and R_2 are connected in parallel, then the total resistance R , measured in ohms (Ω), is given by:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

If R_1 and R_2 are increasing at a rate of $0.3 \text{ } \Omega/\text{s}$, respectively, how fast is R changing when $R_1 = 80 \Omega$ and $R_2 = 100 \Omega$?

Given: $\frac{dR_1}{dt} = \frac{dR_2}{dt} = 0.3 \Omega/\text{s}$

Unknown: $\frac{dR}{dt} \Big|_{R_1=80\Omega, R_2=100\Omega}$

$$R = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

$$\frac{dR}{dt} = -\frac{-\frac{1}{R_1^2} \cdot \frac{dR_1}{dt} - \frac{1}{R_2^2} \cdot \frac{dR_2}{dt}}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)^2}$$

$$\frac{dR}{dt} = \frac{\frac{0.3}{80^2} + \frac{0.3}{100^2}}{\left(\frac{1}{80} + \frac{1}{100}\right)^2}$$

$$\boxed{\frac{dR}{dt} = \frac{41}{270} \Omega/\text{s}}$$

R increases at a rate of $\frac{41}{270} \Omega/\text{s}$.

Problem 18

Two sides of a triangle have lengths 12 m and 15 m. The angle between them is increasing at a rate of $2^\circ/\text{min}$. How fast is the length of the third side increasing when the angle between the sides of fixed length is 60° ?

Given: $\frac{d\theta}{dt} = 2^\circ/\text{min}$, $a = 12$ m, $b = 15$ m

Unknown: $\left. \frac{dx}{dt} \right|_{\theta=60^\circ}$

$$\begin{aligned}
 x^2 &= a^2 + b^2 - 2ab \cos \theta \\
 x &= \sqrt{12^2 + 15^2 - 2 \cdot 12 \cdot 15 \cdot \frac{1}{2}} = 3\sqrt{21} \\
 2x \cdot \frac{dx}{dt} &= 2ab \sin \theta \frac{d\theta}{dt} \\
 \frac{dx}{dt} &= \frac{ab \sin \theta \frac{d\theta}{dt}}{x} = \frac{12 \cdot 15 \cdot \frac{\sqrt{3}}{2} \cdot \frac{\pi}{90}}{3\sqrt{21}} = \boxed{\frac{\pi}{3\sqrt{7}} \text{ m/min}}
 \end{aligned}$$

The length of the third side increases at a rate of $\frac{\pi}{3\sqrt{7}}$ m/min.

Problem 19

The television camera is positioned 4000 ft from the base of a rocket launching pad. Let's assume the rocket rises vertically and its speed is 600 ft/s when it has risen 3000 ft.

(a) How fast is the distance from the television camera to the rocket changing at that moment?

$$\begin{aligned}
 D^2 &= x^2 + y^2 \\
 2d \cdot \frac{dD}{dt} &= 2y \cdot \frac{dy}{dt} \\
 \frac{dD}{dt} &= \frac{2 \cdot 3000 \cdot 600}{2 \cdot 5000} \\
 \boxed{\frac{dD}{dt} = 360 \text{ ft/s}}
 \end{aligned}$$

The distance is increasing at a rate of 360 ft/s.

(b) If the television camera is always kept aimed at the rocket, how fast is the camera's angle of elevation changing at that same moment?

$$\begin{aligned}
 \tan \theta &= \frac{y}{x} \\
 \theta &= \arctan \frac{y}{x} \\
 \frac{d\theta}{dt} &= \frac{1}{1 + (\frac{y}{x})^2} \cdot \frac{dy}{dt} \cdot \frac{1}{x}
 \end{aligned}$$

$$\frac{d\theta}{dt} = \frac{1}{1 + (\frac{3}{4})^2} \cdot \frac{600}{4000}$$

$$\boxed{\frac{d\theta}{dt} = \frac{12}{125} \text{ rads/s}}$$

The camera's angle of elevation is increasing at a rate of $\frac{12}{125}$ rads/s.

Problem 20

A plane flies horizontally at an altitude of 5 km and passes directly over a tracking telescope on the ground. When the angle of elevation is $\pi/3$, this angle is decreasing at a rate of $\pi/6$ rad/min. How fast is the plane traveling at that time?

Given: $h = 5$ km, $\frac{d\theta}{dt} = -\frac{\pi}{6}$ rad/min

Unknown: $\frac{dx}{dt} \Big|_{\theta=\frac{\pi}{3}}$

$$\begin{aligned} \tan \theta &= \frac{h}{x} \implies x = \frac{5}{\sqrt{3}} \\ \sec^2 \theta \cdot \frac{d\theta}{dt} &= h \cdot \left(-\frac{1}{x^2}\right) \cdot \frac{dx}{dt} \\ \frac{dx}{dt} &= -\frac{d\theta}{dt} \cdot \frac{x^2 \sec^2 \theta}{h} = \frac{\pi}{6} \cdot \frac{25}{5 \cdot 3} \cdot 4 = \boxed{\frac{10\pi}{9} \text{ km/min}} \end{aligned}$$

The plane is traveling at a velocity of $\frac{10\pi}{9}$ km/min.

Problem 21

A plane flying with a constant speed of 300 km/h passes over a ground radar station at an altitude of 1km and climbs at an angle of 30° . At what rate is the distance from the plane to the radar station increasing a minute later?

Given: $v = 300$ km/h, $\theta = \frac{\pi}{6}$

Unknown: $\frac{dD}{dt} \Big|_{t=1}$

$$\begin{aligned} D^2 &= x^2 + y^2 \\ D^2 &= \left(\frac{\sqrt{3}}{2} \cdot 5t\right)^2 + \left(1 + \frac{1}{2} \cdot 5t\right)^2 \\ 2D \cdot \frac{dD}{dt} &= 5\sqrt{3}t \cdot \frac{5\sqrt{3}}{2} + (2 + 5t) \cdot \frac{5}{2} \\ \frac{dD}{dt} &= \frac{5\sqrt{3}t \cdot \frac{5\sqrt{3}}{2} + (2 + 5t) \cdot \frac{5}{2}}{2\sqrt{\left(\frac{5\sqrt{3}}{2}\right)^2 + \left(\frac{7}{2}\right)^2}} \end{aligned}$$

$$\boxed{\frac{dD}{dt} = \frac{55}{2\sqrt{31}} \text{ km/min}}$$

The distance from the plane to the radar station increases at a rate of $\frac{55}{2\sqrt{31}}$ km/min.

Problem 22

A runner sprints around a circular track of radius 100 m at a constant speed of 7 m/sec. The runner's friend is standing at a distance 200 m from the center of the track. How fast is the distance between the friends changing when the distance between them is 200 m?

Given: $r = 100$, $\frac{dx}{dt} = 7$

Unknown: $\left. \frac{dD}{dt} \right|_{D=200}$

$$D^2 = x^2 + 200^2 - 2 \cdot 200x \cos \theta$$

$$40000 = 50000 - 40000 \cos \theta \implies \cos \theta = \frac{1}{4}$$

$$2D \cdot \frac{dD}{dt} = 400x \sin \theta \cdot \frac{d\theta}{dt}$$

$$\frac{dx}{dt} = 7 \implies \frac{d\theta}{dt} = \frac{7}{100}$$

$$\frac{dD}{dt} = 100 \cdot \frac{7}{100} \cdot \frac{\sqrt{15}}{4}$$

$$\boxed{\frac{dD}{dt} = \frac{7\sqrt{15}}{4} \text{ m/s}}$$

The distance between the friends are changing at a rate of $\frac{7\sqrt{15}}{4}$ m/s.